

Refine Explanation, Questioning, and Evidence Use

SDA HSF Step 12

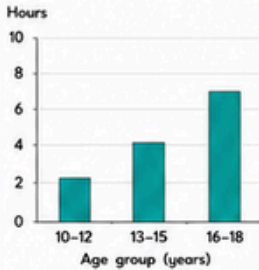
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★ I can use data carefully to ask thoughtful questions, give clear explanations, and support my ideas with strong evidence.

1 Let's Look

Study each data display. Think of questions you could ask.

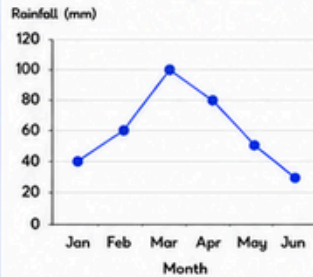
A. Online Time by Age Group (hours per day)



Think of questions:

- _____
- _____
- _____
- _____

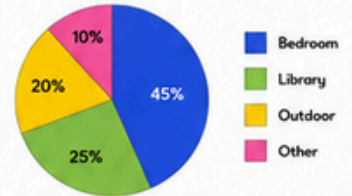
B. Monthly Rainfall (mm)



Think of questions:

- _____
- _____
- _____
- _____

C. Preferred Study Places (survey)



Think of questions:

- _____
- _____

2 Let's Try

Answer the questions using evidence from the data. Explain your thinking.

A. Online Time by Age Group

- Which age group spends the most time online each day?

- How much more time do 16-18 year olds spend online compared to 10-12 year olds?

- Do you think the data shows that online time increases with age? Use evidence to support your answer.

- What other information would help you understand this data better?

B. Monthly Rainfall

- In which month was the rainfall the highest? How much was it?

- Between which two months did the rainfall decrease the most? By how many millimetres?

- What is the overall trend from January to June? Use evidence.

- What question could you ask someone about this data?

C. Preferred Study Places

- What percentage of students prefer the library?

- How many more students prefer the bedroom than outdoor study if 200 students were surveyed?

- Is outdoor study more popular than other places combined? Explain using the data.

- What does this data tell us about where students like to study?

3 Let's Do

Read the scenario and complete the tasks.

A local council surveyed 600 residents about community transport.

How People Currently Travel

Transport	Number of people	Percentage (%)
Car	270	45%
Public transport	150	25%
Walk or cycle	120	20%
Other	60	10%
Total	600	100%

Residents' Satisfaction (out of 5)

Mean (average)	3.1
Median	3
Mode	3
Range	1 to 5

- What does the mode of 3 tell us about satisfaction?

- Is the mean higher or lower than the median? What might this suggest?

- Are more people using public transport or walk/cycle? By how many people?

- If the council wants more people to use public transport, what question should they ask residents?

- Write a claim the council could make to support improving public transport.

- What evidence from the data supports your claim in part (e)?

Think About It

- What other data might help the council make a better decision?

- How could the way the question was asked affect the results?

4 Challenge

Choose a real-world context and complete the task.

Choose a context you are interested in.

- Examples:
- Screen time and wellbeing
 - Recycling in your school
 - Sports participation
 - Water usage at home

Create your data display.

Choose one:

- ☐ Bar chart
- ☐ Line graph
- ☐ Pie chart
- ☐ Table

Answer and Explain

- What is one interesting thing your data shows?

- What question would you ask someone about your data?

- What claim can you make based on your data? Use evidence to support your claim.

Evaluate

- How reliable is your data? Why?

- What could make your survey or data better?

My Learning Check

How did I go today?



I can do it!



Almost there!



I need more help.



This was hard.

Circle a face.

Parent Observation (optional)

Tick what you saw today.

- ☐ Asked thoughtful questions about data
- ☐ Used evidence to support ideas
- ☐ Explained thinking clearly
- ☐ Connected data to real situations
- ☐ Reflected on the quality of data
- ☐ Needed support

Notes: _____

Key Words

data – facts and information collected
evidence – information used to support an idea
claim – a statement or conclusion
explain – make an idea clear with reasons
question – something you ask to find out more
trend – a pattern or change over time
mean – the average of a set of values
median – the middle value when ordered
mode – the most common value
range – the difference between the highest and lowest values